```
# STEP 1: IMPORT LIBRARIES
import numpy as np
import pandas as pd
from sklearn import svm
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(font_scale=1.2)


# STEP 2: LOAD DATASET (Upload in Colab)
from google.colab import files
uploaded = files.upload()

recipes = pd.read_csv('recipes_muffins_cupcakes.csv')

print(recipes.head())
print(recipes.shape)


# STEP 3: VISUALIZE DATA
sns.lmplot(x='Sugar', y='Flour', data=recipes, hue='Type',
           palette='Set1', fit_reg=False, scatter_kws={"s":70})


# STEP 4: PREPARE DATA
sugar_flour = recipes[['Sugar', 'Flour']].values
type_label = np.where(recipes['Type'] == 'Muffin', 0, 1)


# STEP 5: TRAIN SVM MODEL
model = svm.SVC(kernel='linear')
model.fit(sugar_flour, type_label)


# STEP 6: FIND HYPERPLANE
w = model.coef_[0]
a = -w[0] / w[1]

xx = np.linspace(5, 30)
yy = a * xx - (model.intercept_[0] / w[1])


# STEP 7: MARGIN LINES
b = model.support_vectors_[0]
yy_down = a * xx + (b[1] - a * b[0])

b = model.support_vectors_[-1]
yy_up = a * xx + (b[1] - a * b[0])


# STEP 8: PLOT HYPERPLANE
sns.lmplot(x='Sugar', y='Flour', data=recipes, hue='Type',
           palette='Set1', fit_reg=False, scatter_kws={"s":70})

plt.plot(xx, yy, linewidth=2, color='black')
plt.plot(xx, yy_down, 'k--')
plt.plot(xx, yy_up, 'k--')

plt.scatter(model.support_vectors_[:, 0],
            model.support_vectors_[:, 1],
            s=80, facecolors='none')

plt.show()


# STEP 9: TRAIN TEST SPLIT
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    sugar_flour, type_label, test_size=0.2
)


# STEP 10: TRAIN NEW MODEL
model1 = svm.SVC(kernel='linear')
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model1.fit(x_train, y_train)


# STEP 11: PREDICTION
pred = model1.predict(x_test)
print(pred)


# STEP 12: EVALUATION
from sklearn.metrics import confusion_matrix, classification_report

print(confusion_matrix(y_test, pred))
print(classification_report(y_test, pred))


# STEP 13: RESULT
print("\nRESULT:")
print("Thus the python program to implement SVM classifier model has been executed successfully and the classified output has bee


print('\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n')
print('\n\n\n\n\n\n\n\n\n\n\n\n\n\n')
```
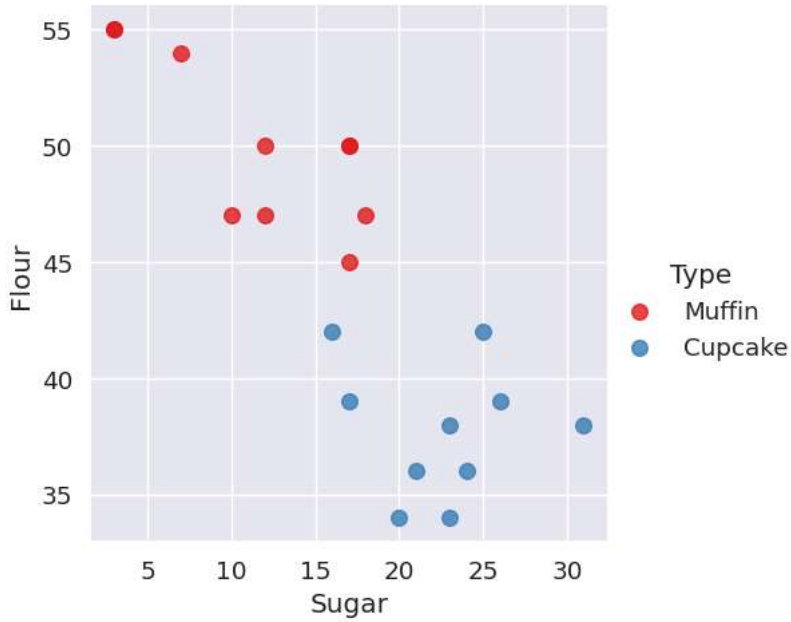
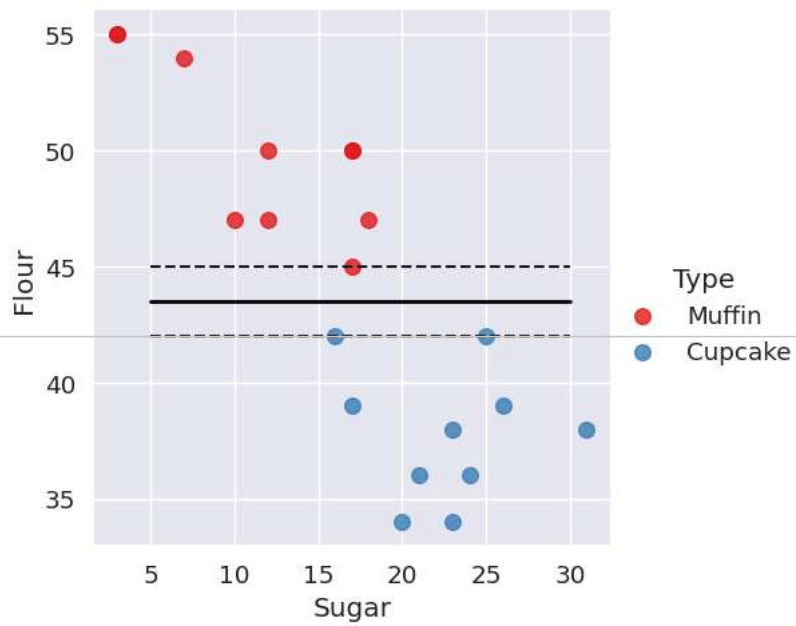

Choose Files recipes_mu…pcakes.csv

recipes_muffins_cupcakes.csv(text/csv) - 622 bytes, last modified: 4/11/2026 - 100% done

```
Saving recipes_muffins_cupcakes.csv to recipes_muffins_cupcakes (2).csv
     Type  Flour  Milk  Sugar  Butter  Egg  Baking Powder  Vanilla  Salt
0  Muffin     55    28      3       7    5              2        0     0
1  Muffin     47    24     12       6    9              1        0     0
2  Muffin     47    23     18       6    4              1        0     0
3  Muffin     45    11     17      17    8              1        0     0
4  Muffin     50    25     12       6    5              2        1     0
(20, 9)
```






```
[0 1 0 0]
[[3 0]
 [0 1]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         1

    accuracy                           1.00         4
   macro avg       1.00      1.00      1.00         4
weighted avg       1.00      1.00      1.00         4
```